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Fundamental Concepts of Data Security

Secret-Key Encryption

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Outline

- Substitution Technique
  - Ceasar Cipher
  - Monoalphabetic substitution cipher
- DES and AES Encryption Algorithms
- Encryption Modes

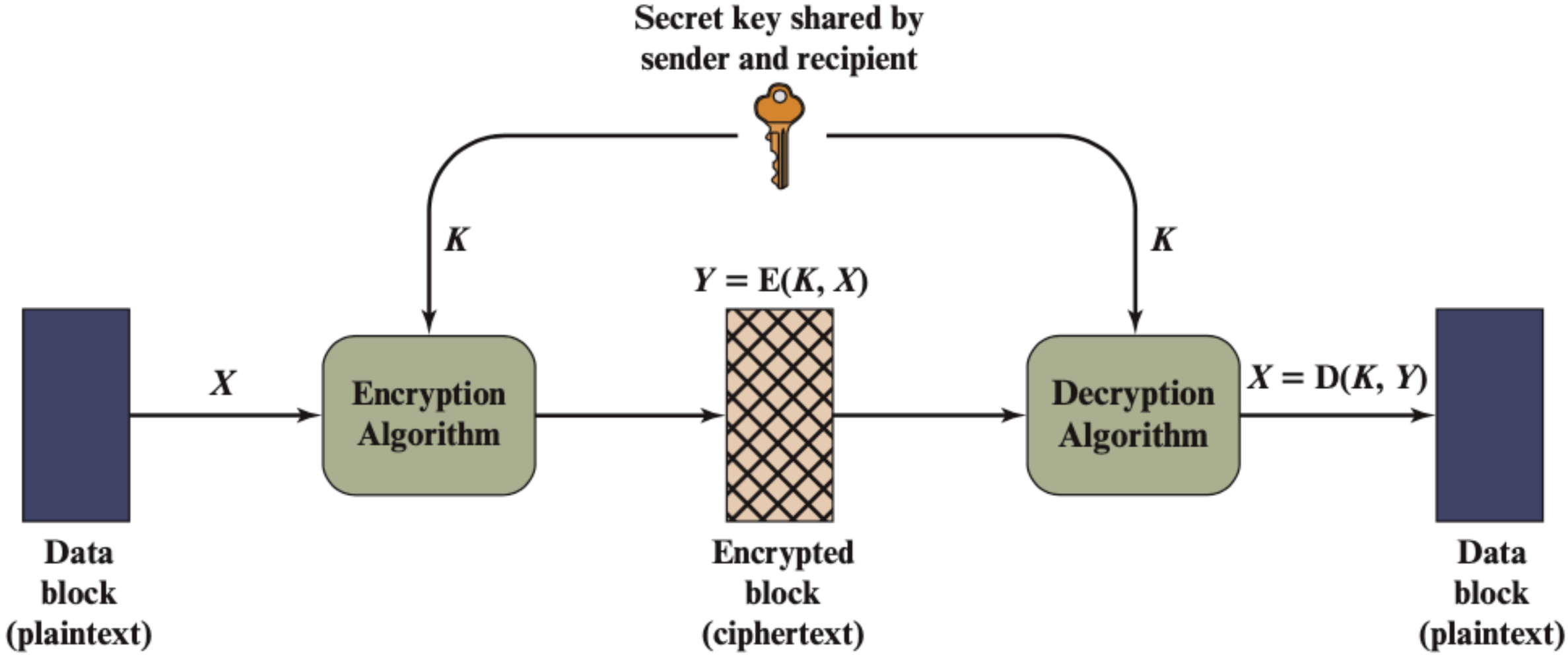


Figure 3.1 Simplified Model of Symmetric Encryption

Bruteforce attack → longer the key, more secure the cipher, provided the algorithm is secure.

Cryptanalysis and Brute-Force Attack

- **Cryptanalysis**: attacks rely on the nature of the algorithm plus perhaps some knowledge of the general characteristics of the plaintext or even some sample plaintext–ciphertext pairs
- **Brute-force attack**: tries every possible key on a piece of cipher- text

Security of Encryption Scheme

- **unconditionally secure**: no matter how much time an opponent has, it is impossible for him or her to decrypt the ciphertext simply because the required information is not there
- **computationally secure**: the time or cost of breaking the cipher exceeds the useful lifetime or the value of the encrypted information

Substitution Techniques

Ceasar Cipher

The earliest known, and the simplest, use of a substitution cipher was by Julius Caesar.

```
plain:  a b c d e f g h i j k l m n o p q r s t u v w x y z
cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C
```

For example,

```
plain:  meet me after the toga party
cipher: PHHW PH DIWHU WKH WRJD SDUWB
```

For each plaintext letter p, substi- tute the ciphertext letter C

$$C = \mathrm{E}(3, p) = (p + 3) \bmod 26$$

The general Caesar algorithm is

$$C = \mathrm{E}(k, p) = (p + k) \bmod 26$$

Brute-force on Ceasar Cipher

KEY	PHHW PH DIWHU WKH WRJD SDUWB
1	oggv og chvgt vjg vqic rctva
2	nffu nf bgufs uif uphb qbsuz
3	meet me after the toga party
4	ldds ld zesdq sgd snfz ozqsx
5	kccr kc ydrcp rfc rmey nyprw
6	jbbq jb xcqbo qeb qldx mxoqv
7	iaap ia wbpan pda pkcw lwnpu
8	hzzo hz vaozm ocz ojbv kvmot
9	gyyn gy uznyl nby niau julns
10	fxxm fx tymxk max mhzt itkmr
11	ewwl ew sxlwj lzw lgys hsjlq
12	dvvk dv rwkvi kyv kfxr grikp
13	cuuj cu qvjuh jxu jewq fqhjo
14	btti bt puitg iwt idvp epgin
15	assh as othsf hvs hcuo dofhm
16	zrrg zr nsgre gur gbtn cnegl
17	yqqf yq mrfqd ftq fasm bmdfk
18	xppe xp lqepc esp ezrl alcej
19	wood wo kpdob dro dyqk zkbdi
20	vnnc vn jocna cqn cxpj yjach
21	ummb um inbmz bpm bwoi xizbg
22	tlla tl hmaly aol avnh whyaf
23	skkz sk glzkx znk zumg vgxze
24	rjjy rj fkyjw ymj ytlf ufwyd
25	qiix qi ejxiv xli xske tevxc

Figure 3.3 Brute-Force Cryptanalysis of Caesar Cipher

Monoalphabetic substitution cipher

- **Substitution ciphers** are probably the most common form of cipher. They work by replacing each letter of the plaintext (and sometimes puntuation marks and spaces) with another letter (or possibly even a random symbol).

- A **monoalphabetic substitution cipher**, also known as a simple substitution cipher, relies on a **fixed replacement** structure. That is, the substitution is fixed for each letter of the alphabet. Thus, if "a" is encrypted to "R", then every time we see the letter "a" in the plaintext, we replace it with the letter "R" in the ciphertext.

```
plaintext alphabet = abcdefghijklmnopqrstuvwxyz
key alphabet       = isrvhfuwceaqjxpygdbtlokmzn
```

article.txt

Star Trek's Captain Kirk rocketing into space next week

Jeff Bezos' space travel company, Blue Origin, announced Monday that William Shatner will blast off from West Texas on Oct. 12.

At age 90, Shatner will become the oldest person in space. He'll join three others — two of them paying customers — aboard a Blue Origin capsule. It will be the company's second launch with a crew.

Bezos was on the debut flight in July, along with his brother and the youngest and oldest to fly in space. Shatner will break that upper threshold by six years.

"I've heard about space for a long time now. I'm taking the opportunity to see it for myself. What a miracle," Shatner said in a statement.

Also flying with Shatner: a former NASA engineer who founded a nanosatellite company; the co-founder of a software company specializing in clinical research; and a Blue Origin employee. The up-and-down space hop will last 10 minutes and reach no higher than about 66 miles (106 kilometers).

Change all the upper-case letters to lower-case ones

seed@VM:~/.../Lab1$ **tr [:upper:] [:lower:] < article.txt > lowercase.txt**
seed@VM:~/.../Lab1$ **cat lowercase.txt**
star trek's captain kirk rocketing into space next week

jeff bezos' space travel company, blue origin, announced monday that william shatner will blast off from west texas on oct. 12.

at age 90, shatner will become the oldest person in space. he'll join three others — two of them paying customers — aboard a blue origin capsule. it will be the company's second launch with a crew.

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Remove numbers and special characters

seed@VM:~/.../Lab1$ **tr -cd '[a-z][\n][:space:]' < lowercase.txt > plaintext.txt**

seed@VM:~/.../Lab1$ **cat plaintext.txt**

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Generate a random key → fixed mapping

```
#!/bin/env python3
import random


s = ""
# Generate string a..z
for chCode in range(ord('a'), ord('z') + 1):
        s += chr(chCode)
lst  = random.sample(s, len(s))
key = ''.join(lst)
print("plaintext alphabet = {}".format(s))
print("key alphabet       = {}".format(key))
```

```
seed@VM:~/.../Lab1$ cat key.py
#!/bin/env python3
import random
s = ""
# Generate string a..z
for chCode in range(ord('a'), ord('z') + 1):
                s += chr(chCode)
lst  = random.sample(s, len(s))
key = ''.join(lst)
print("plaintext alphabet = {}".format(s))
print("key alphabet       = {}".format(key))
seed@VM:~/.../Lab1$ python key.py
plaintext alphabet = abcdefghijklmnopqrstuvwxyz
key alphabet       = wjmtxzorvqdukfysnlhipabcge
```

Encryption

seed@VM:~/.../Lab1$ **python key.py**

plaintext alphabet = abcdefghijklmnopqrstuvwxyz

key alphabet = wjmtxzorvqdukfysnlhipabcge

seed@VM:~/.../Lab1$ **tr 'a-z' 'wjmtxzorvqdukfysnlhipabcge' < plaintext.txt > ciphertext.txt**

seed@VM:~/.../Lab1$ **cat ciphertext.txt**

hiwl ilxdh mwsiwvf dvld lymdxivfo vfiy hswmx fxci bxxd

qxzz jxeyh hswmx ilwaxu mykswfg jupx ylvovf wffypfmxt kyftwg irwi bvuuvwk hrwifxl bvuu juwhi yzz zlyk bxhi ixcwh yf ymi

wi wox hrwifxl bvuu jxmykx irx yutxhi sxlhyf vf hswmx rxuu qyvf irlxx yirxlh iby yz irxk swgvfo mphiykxlh wjywlt w jupx ylvovf mwshpux vi bvuu jx irx mykswfgh hxmyft uwpfmr bvir w mlxb

jxeyh bwh yf irx txjpi zuvori vf qpug wuyfo bvir rvh jlyirxl wft irx gypfoxhi wft yutxhi iy zug vf hswmx hrwifxl bvuu jlxwd irwi pssxl irlxhryut jg hvc gxwlh

vax rxwlt wjypi hswmx zyl w uyfo ivkx fyb vk iwdvfo irx yssylipfvig iy hxx vi zyl kghxuz brwi w kvlwmux hrwifxl hwvt vf w hiwixkxfi

wuhy zugvfo bvir hrwifxl w zylkxl fwhw xfovfxxl bry zypftxt w fwfyhwixuuvix mykswfg irx myzypftxl yz w hyzibwlx mykswfg hsxmvwuvevfo vf muvfvmwu lxhxwlmr wft w jupx ylvovf xksuygxx irx pswfttybf hswmx rys bvuu uwhi kvfpixh wft lxwmr fy rvorxl irwf wjypi kvuxh dvuykxixlh

Decrypt

seed@VM:~/.../Lab1$ **tr 'a-z' 'wjmtxzorvqdukfysnlhipabcge' < plaintext.txt > ciphertext.txt**

seed@VM:~/.../Lab1$ **tr 'wjmtxzorvqdukfysnlhipabcge' 'a-z' < ciphertext.txt**

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Breaking Monoalphabetic Substitution Cipher

Exploit the regularities of the language

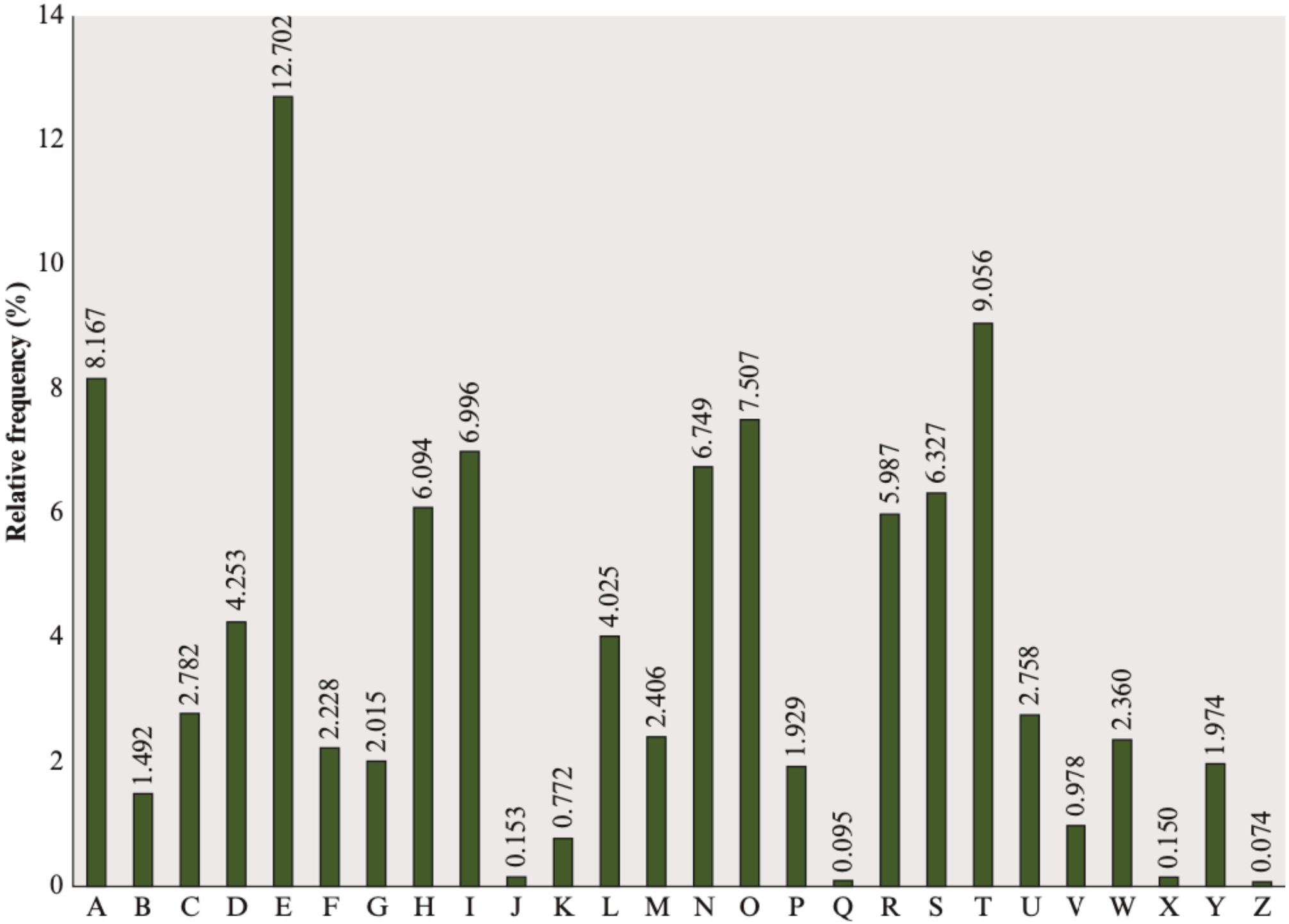


Figure 3.5 Relative Frequency of Letters in English Text

Counter/Analyzer (dcode.fr)

Letters by frequency of appearance in English:

E	12.7 %	**T**	9.1 %	**A**	8.2 %
O	7.5 %	**I**	7.0 %	**N**	6.7 %
S	6.3 %	**H**	6.1 %	**R**	6.0 %
L	4.0 %	**D**	4.3 %	**C**	2.8 %
U	2.8 %	**M**	2.4 %	**W**	2.4 %
F	2.2 %	**G**	2.0 %	**Y**	2.0 %
P	1.9 %	**B**	1.5 %	**V**	1.0 %
K	0.8 %	**J**	0.2 %	**X**	0.2 %
Q	0.1 %	**Z**	0.1 %		

Occurency and Frequency Analysis

1-grams

% calculated | % expected

↑↓	↑↓	↑↓	↑↓
X	86×	11.33%	
W	71×	9.35%	
I	64×	8.43%	
F	58×	7.64%	
Y	57×	7.51%	
V	51×	6.72%	
H	47×	6.19%	
U	42×	5.53%	
L	41×	5.4%	
R	36×	4.74%	
M	27×	3.56%	
S	22×	2.9%	
K	20×	2.64%	
B	19×	2.5%	
T	19×	2.5%	
P	18×	2.37%	
Z	17×	2.24%	
O	15×	1.98%	
J	15×	1.98%	
G	15×	1.98%	
D	8×	1.05%	
Q	3×	0.4%	
E	3×	0.4%	
C	3×	0.4%	
A	2×	0.26%	
#N : 25	Σ = 759.00	Σ = 100.00	#N : 25

[Bigram - Wikipedia]

Bigram frequency in the English language [edit]

The frequency of the most common letter bigrams in a small English corpus is:[4]

th 1.52	en 0.55	ng 0.18
he 1.28	ed 0.53	of 0.16
in 0.94	to 0.52	al 0.09
er 0.94	it 0.50	de 0.09
an 0.82	ou 0.50	se 0.08
re 0.68	ea 0.47	le 0.08
nd 0.63	hi 0.46	sa 0.06
at 0.59	is 0.46	si 0.05
on 0.57	or 0.43	ar 0.04
nt 0.56	ti 0.34	ve 0.04
ha 0.56	as 0.33	ra 0.04
es 0.56	te 0.27	ld 0.02
st 0.55	et 0.19	ur 0.02

Occurency and Frequency Analysis
2-grams

↑↓	↑↓	↑↓	↑↓
IR	12×	3.17%	
XL	11×	2.9%	
VF	10×	2.64%	
HI	7×	1.85%	
SW	7×	1.85%	
UU	7×	1.85%	
BV	7×	1.85%	
YZ	6×	1.58%	
RW	6×	1.58%	
YL	6×	1.58%	
WL	6×	1.58%	
WM	6×	1.58%	
RX	6×	1.58%	
YK	5×	1.32%	
FT	5×	1.32%	
LX	5×	1.32%	
WF	5×	1.32%	
FO	5×	1.32%	
WI	5×	1.32%	
HX	4×	1.06%	
TW	4×	1.06%	
XM	4×	1.06%	
IF	4×	1.06%	
FW	4×	1.06%	

Letter Ngram Counter/Analyzer (dcode.fr)

Frequency Analysis (advanced)

TEXT TO ANALYZE

```
WUHY ZUGVFO BVIR HRWIFXL W ZYLKXL FWHW XFOVFXXL BRY ZYPFT
XT W FWFYHWIXUUVIX MYKSWFG IRX MYZYPFTXL YZ W HYZIBWLX MY
KSWFG HSXMVWUVEVFO VF MUVFVMWU LXHXWLMR WFT W JUPX YLVOVF
XKSUYGXX IRX PSWFTTYBF HSWMX RYS BVUU UWHI  KVFPIXH WFT L
XWMR FY RVORXL IRWF WJYPI  KVUXH  DVUYKXIXLH
```

PLAINTEXT EXPECTED LANGUAGE English

Target characters for frequency analysis

- (•) LETTERS (A-Z) ONLY
- ( ) LETTERS (A-Z) AND DIGITS (0-9) ONLY
- ( ) DIGITS (0-9) ONLY
- ( ) ONLY THESE CHARACTERS: αβγδε
- ( ) ALL CHARACTERS (INCLUDING PUNCTUATION AND SYMBOLS)

STANDARDIZE LETTERS (IGNORE UPPER-LOWER CASE AND DIACRITICS) [x]

Items to Analyze

- ( ) EACH CHARACTER SEPARATELY
- (•) BIGRAMS (COUPLES OF 2 CHARACTERS)
- ( ) TRIGRAMS (SET OF 3 CHARACTERS)
- ( ) N-GRAMS N= 4



Listing 21.1: Bigram and trigram frequencies

```
Bigram frequency in English
---------------------------------------------------------
TH :  2.71        EN :  1.13        NG :  0.89
HE :  2.33        AT :  1.12        AL :  0.88
IN :  2.03        ED :  1.08        IT :  0.88
ER :  1.78        ND :  1.07        AS :  0.87
AN :  1.61        TO :  1.07        IS :  0.86
RE :  1.41        OR :  1.06        HA :  0.83
ES :  1.32        EA :  1.00        ET :  0.76
ON :  1.32        TI :  0.99        SE :  0.73
ST :  1.25        AR :  0.98        OU :  0.72
NT :  1.17        TE :  0.98        OF :  0.71

Bigram frequency in chiphertext (The top-10 patterns)
---------------------------------------------------------
tn  :  77         np  :  50
yt  :  76         hn  :  45
nh  :  61         nu  :  44
nq  :  51         mu  :  42
vu  :  51         cv  :  42


Trigram frequency in English
---------------------------------------------------------
THE :  1.81       ERE :  0.31       HES :  0.24
AND :  0.73       TIO :  0.31       VER :  0.24
ING :  0.72       TER :  0.30       HIS :  0.24
ENT :  0.42       EST :  0.28       OFT :  0.22
```

Occurency and Frequency Analysis 3-grams

↑↓	↑↓	↑↓	↑↓
IRX	4×	1.58%	
HSW	3×	1.19%	
WIF	3×	1.19%	
VUU	3×	1.19%	
KSW	3×	1.19%	
XHI	3×	1.19%	
XMY	3×	1.19%	
FOV	3×	1.19%	
XHR	2×	0.79%	
XLB	2×	0.79%	
FGH	2×	0.79%	
UUJ	2×	0.79%	
HIW	2×	0.79%	
VOV	2×	0.79%	
YKX	2×	0.79%	
VFW	2×	0.79%	
YUT	2×	0.79%	
XYL	2×	0.79%	
PFT	2×	0.79%	
YZY	2×	0.79%	
JUP	2×	0.79%	

Frequency Analysis

Cryptography › Cryptanalysis › Frequency Analysis

Frequency Analysis (advanced)

TEXT TO ANALYZE

```
WUHY ZUGVFO BVIR HRWIFXL W ZYLKXL FWHW XFOVFXXL BRY ZYPFT
XT W FWFYHWIXUUVIX MYKSWFG IRX MYZYPFTXL YZ W HYZIBWLX MY
KSWFG HSXMVWUVEVFO VF MUVFVMWU LXHXWLMR WFT W JUPX YLVOVF
XKSUYGXX IRX PSWFTTYBF HSWMX RYS BVUU UWHI  KVFPIXH WFT L
XWMR FY RVORXL IRWF WJYPI  KVUXH  DVUYKXIXLH
```

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- ( ) ONLY THESE CHARACTERS: αβγδε
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- STANDARDIZE LETTERS (IGNORE UPPER-LOWER CASE AND DIACRITICS) [x]

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- (•) TRIGRAMS (SET OF 3 CHARACTERS)
- ( ) N-GRAMS N= 4

(FOR NGRAMS)
- (•) BLOCKS ANALYSIS (ABCDEF => AB,CD,EF)
- ( ) SLIDING WINDOW/OVERLAPPING (ABCDEF => AB,BC,CD,DE,EF)

KEEP WORDS BORDERS (ABC_DE ≠ ABCDE) [ ]

Guess work:
x → E
i → T
r → H

seed@VM:~/.../Lab1$ **tr 'irx' 'THE' < ciphertext.txt**

hTwl TlEdh mwsTwvf dvld lymdETvfo vfTy hswmE fEcT bEEd

qEzz jEeyh hswmE TlwaEu mykswfg jupE ylvovf wffypfmEt kyftwg THwT bvuuvwk hHwTfEl bvuu juwhT yzz zlyk bEhT TEcwh yf ymT

wT woE hHwTfEl bvuu jEmykE THE yutEhT sElhyf vf hswmE HEuu qyvf THlEE yTHElh Tby yz THEk swgvfo mphTykElh wjywlt w jupE ylvovf mwshpuE vT bvuu jE THE mykswfgh hEmyft uwpfmH bvTH w mlEb

jEeyh bwh yf THE tEjpT zuvoHT vf qpug wuyfo bvTH Hvh jlyTHEl wft THE gypfoEhT wft yutEhT Ty zug vf hswmE hHwTfEl bvuu jlEwd THwT pssEl THlEhHyut jg hvc gEwlh

vaE HEwlt wjypT hswmE zyl w uyfo TvkE fyb vk Twdvfo THE yssylTpfvTg Ty hEE vT zyl kghEuz bHwT w kvlwmuE hHwTfEl hwvt vf w hTwTEkEfT

wuhy zugvfo bvTH hHwTfEl w zylkEl fwhw EfovfEEl bHy zypftEt w fwfyhwTEuuvTE mykswfg THE myzypftEl yz w hyzTbwlE mykswfg hsEmvwuvevfo vf muvfvmwu lEhEwlmH wft w jupE ylvovf EksuygEE THE pswfttybf hswmE Hys bvuu uwhT kvfpTEh wft lEwmH fy HvoHEl THwf wjypT kvuEh dvuykETElh

Guess next:
w → A
l → R (xl ← er)
v → l
f→ N

seed@VM:~/.../Lab1$ **tr 'irxwlvf' 'THEARIN' < ciphertext.txt**

hTAR TREdh mAsTAIN dIRd RymdETINo INTy hsAmE NEcT bEEd

qEzz jEeyh hsAmE TRAaEu myksANg jupE yRIoIN ANNypNmEt kyNtAg THAT bIuuIAk hHATNER bIuu juAhT yzz zRyk bEhT TEcAh yN ymT

AT AoE hHATNER bIuu jEmykE THE yutEhT sERhyN IN hsAmE HEuu qyIN THREE yTHERh Tby yz THEk sAgINo mphTykERh AjyARt A jupE yRIoIN mAshpuE IT bIuu jE THE myksANgh hEmyNt uApNmH bITH A mREb

jEeyh bAh yN THE tEjpT zuIoHT IN qpug AuyNo bITH HIh jRyTHER ANt THE gypNoEhT ANt yutEhT Ty zug IN hsAmE hHATNER bIuu jREAd THAT pssER THREhHyut jg hIc gEARh

IaE HEARt AjypT hsAmE zyR A uyNo TIkE Nyb Ik TAdINo THE yssyRTpNITg Ty hEE IT zyR kghEuz bHAT A kIRAmuE hHATNER hAIt IN A hTATEkENT

Auhy zugINo bITH hHATNER A zyRkER NAhA ENoINEER bHy zypNtEt A NANyhATEuuITE myksANg THE myzypNtER yz A hyzTbARE myksANg hsEmIAuIeINo IN muINImAu REhEARmH ANt A jupE yRIoIN EksuygEE THE psANttybN hsAmE Hys bIuu uAhT kINpTEh ANt REAmH Ny HIoHER THAN AjypT kIuEh dIuykETERh

seed@VM:~/.../Lab1$ **tr 'irxwlvfhyjtkgsobumpqzedca' 'THEARINSOBDMYPGWLCUJFZKXV' < ciphertext.txt**
STAR TREKS CAPTAIN KIRK ROCKETING INTO SPACE NEXT WEEK

JEFF BEZOS SPACE TRAVEL COMPANY BLUE ORIGIN ANNOUNCED MONDAY THAT WILLIAM SHATNER WILL BLAST OFF FROM WEST TEXAS ON OCT

AT AGE SHATNER WILL BECOME THE OLDEST PERSON IN SPACE HELL JOIN THREE OTHERS TWO OF THEM PAYING CUSTOMERS ABOARD A BLUE ORIGIN CAPSULE IT WILL BE THE COMPANYS SECOND LAUNCH WITH A CREW

BEZOS WAS ON THE DEBUT FLIGHT IN JULY ALONG WITH HIS BROTHER AND THE YOUNGEST AND OLDEST TO FLY IN SPACE SHATNER WILL BREAK THAT UPPER THRESHOLD BY SIX YEARS

IVE HEARD ABOUT SPACE FOR A LONG TIME NOW IM TAKING THE OPPORTUNITY TO SEE IT FOR MYSELF WHAT A MIRACLE SHATNER SAID IN A STATEMENT

ALSO FLYING WITH SHATNER A FORMER NASA ENGINEER WHO FOUNDED A NANOSATELLITE COMPANY THE COFOUNDER OF A SOFTWARE COMPANY SPECIALIZING IN CLINICAL RESEARCH AND A BLUE ORIGIN EMPLOYEE THE UPANDDOWN SPACE HOP WILL LAST MINUTES AND REACH NO HIGHER THAN ABOUT MILES KILOMETERS

Removing the nature of plaintext in ciphertext

- Two principal methods are used in substitution ciphers to lessen the extent to which the structure of the plaintext survives in the ciphertext:
  - (1) to encrypt multiple letters of plaintext
    - e.g., Playfair cipher, Hill cipher
  - (2) to use multiple cipher alphabets
    - Vigenere cipher <— polyalphabetic substitution cipher

One-Time Pad

- The key is to be used to encrypt and decrypt *a single message*, and then is discarded
- Produces random output that bears no statistical relationship to the plaintext
- There is simply **no way** to break the code
- The only cryptosystem that exhibits what is referred to as ***perfect secrecy***

Other Technique

- Transposition technique: performing some sort of permutation on the plaintext letters
  - E.g, rail fence technique

```
m e m a t r h t g p r y
  e t e f e t e o a a t
```

The encrypted message is

```
MEMATRHTGPRYETEFETEOAAT
```

Or

```
Key:          4 3 1 2 5 6 7
Plaintext:    a t t a c k p
              o s t p o n e
              d u n t i l t
              w o a m x y z
Ciphertext:   TTNAAPTMTSUOAODWCOIXKNLYPETZ
```

DES and AES Encryption Algorithms

Stream Ciphers vs Block Ciphers

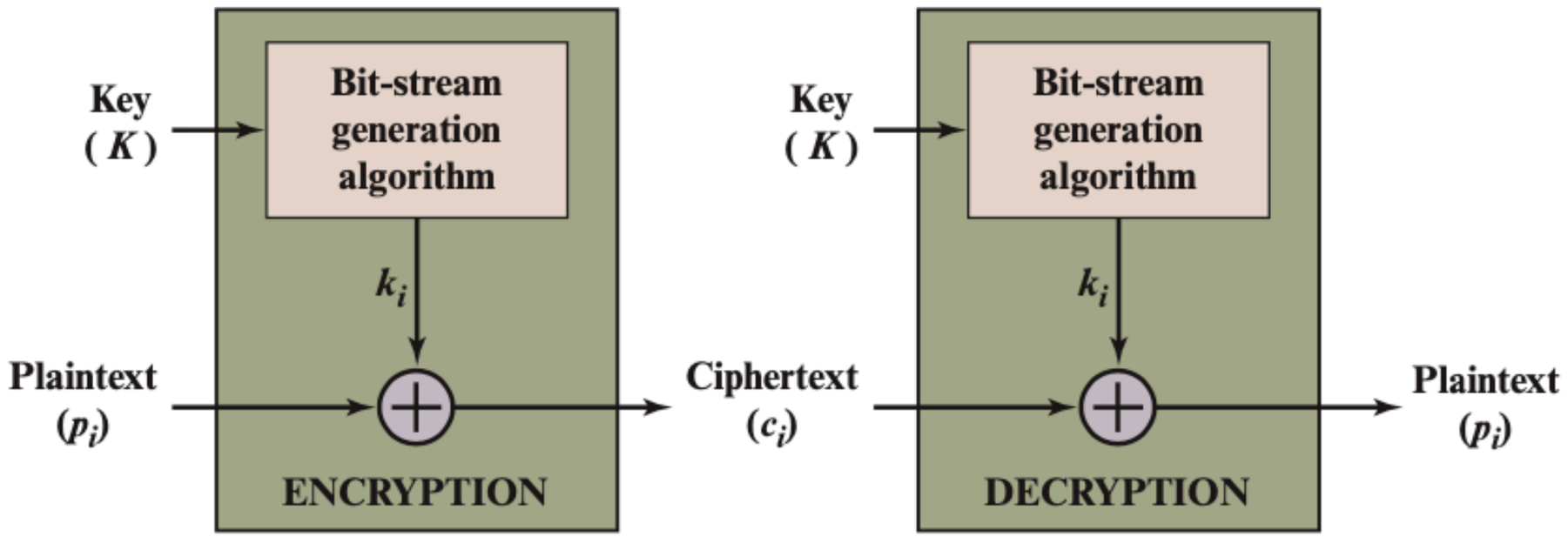


Figure 4.1 Stream Cipher and Block Cipher

Block Ciphers - DES and AES Encryption Algorithms

- Data Encryption Standard (DES) ← 1977, FIPS 46-3, Data Encryption Standard (DES) (withdrawn May 19, 2005) (nist.gov)
  - 56-bit key
  - block size = 64 bits
  - Triple-DES → 168-bit key
- Advanced Encryption Standard (AES) ← 2001, FIPS 197, Advanced Encryption Standard (AES) (nist.gov)
  - key sizes: 128, 192, and 256 bits
  - block size = 128 bits

The Strength of DES and AES

Table 4.5 Average Time Required for Exhaustive Key Search

Key Size (bits)	Cipher	Number of Alternative Keys	Time Required at 10^9 Decryptions/s	Time Required at 10^{13} Decryptions/s
56	DES	$2^{56} \approx 7.2 \times 10^{16}$	2^{55} ns = 1.125 years	1 hour
128	AES	$2^{128} \approx 3.4 \times 10^{38}$	2^{127} ns = 5.3×10^{21} years	5.3×10^{17} years
168	Triple DES	$2^{168} \approx 3.7 \times 10^{50}$	2^{167} ns = 5.8×10^{33} years	5.8×10^{29} years
192	AES	$2^{192} \approx 6.3 \times 10^{57}$	2^{191} ns = 9.8×10^{40} years	9.8×10^{36} years
256	AES	$2^{256} \approx 1.2 \times 10^{77}$	2^{255} ns = 1.8×10^{60} years	1.8×10^{56} years
26 characters (permutation)	Monoalphabetic	$2! = 4 \times 10^{26}$	2×10^{26} ns = 6.3×10^{9} years	6.3×10^{6} years

Which key size?

Table 7.10
Security Requirements for Different Information

Type of Traffic	Lifetime	Minimum Key Length
Tactical military information	minutes/hours	56–64 bits
Product announcements, mergers, interest rates	days/weeks	64 bits
Long-term business plans	years	64 bits
Trade secrets (e.g., recipe for Coca-Cola)	decades	112 bits
H-bomb secrets	>40 years	128 bits
Identities of spies	>50 years	128 bits
Personal affairs	>50 years	128 bits
Diplomatic embarrassments	>65 years	at least 128 bits
U.S. census data	100 years	at least 128 bits

Key length for Public-key

Table 7.9
Symmetric and Public-key Key Lengths with Similar Resistances to Brute-Force Attacks

Symmetric Key Length	Public-key Key Length
56 bits	384 bits
64 bits	512 bits
80 bits	768 bits
112 bits	1792 bits
128 bits	2304 bits

- The size of a typical plaintext is usually larger than the block size, so how do we encrypt them?

Encryption Modes

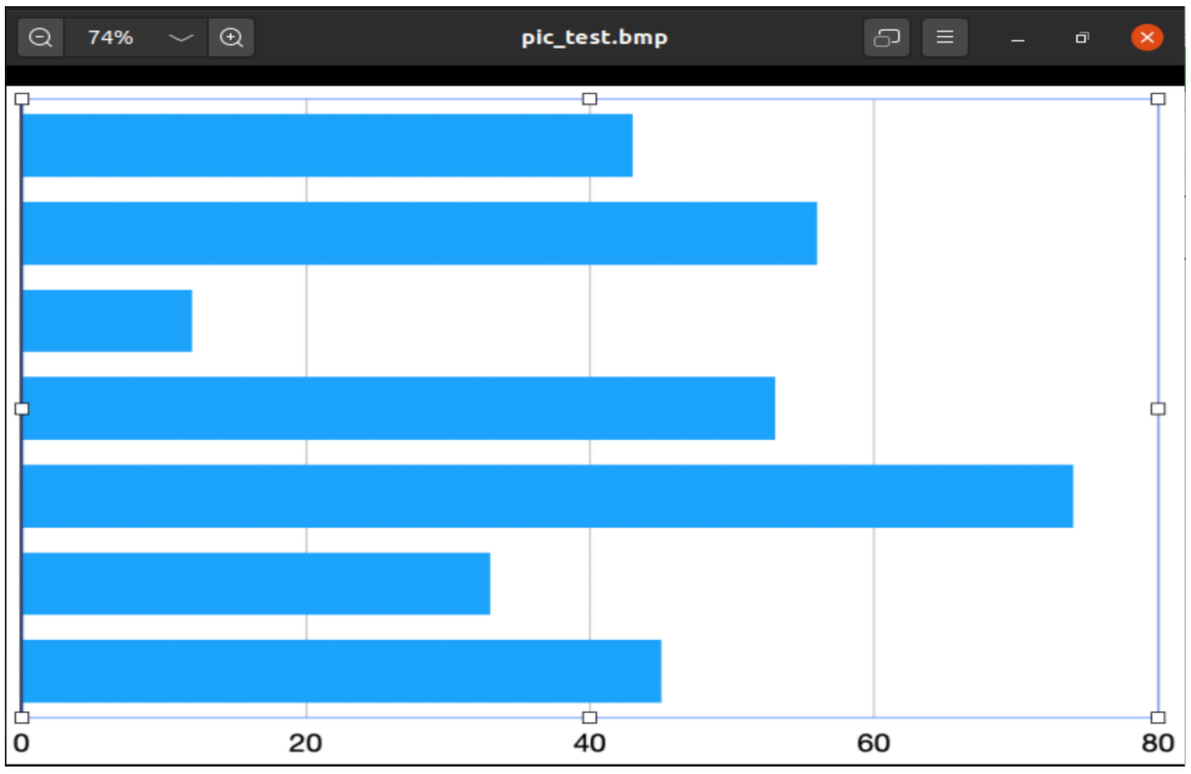
openssl enc

seed@VM:~/.../Lab1$ **openssl enc -ciphers -list**

Supported ciphers:

-aes-128-cbc -aes-128-cfb -aes-128-cfb1
-aes-128-cfb8 -aes-128-ctr -aes-128-ecb
-aes-128-ofb -aes-192-cbc -aes-192-cfb
-aes-192-cfb1 -aes-192-cfb8 -aes-192-ctr
-aes-192-ecb -aes-192-ofb -aes-256-cbc
-aes-256-cfb -aes-256-cfb1 -aes-256-cfb8
-aes-256-ctr -aes-256-ecb -aes-256-ofb
-aes128 -aes128-wrap -aes192
-aes192-wrap -aes256 -aes256-wrap
-aria-128-cbc -aria-128-cfb -aria-128-cfb1
-aria-128-cfb8 -aria-128-ctr -aria-128-ecb
-aria-128-ofb -aria-192-cbc -aria-192-cfb
-aria-192-cfb1 -aria-192-cfb8 -aria-192-ctr
-aria-192-ecb -aria-192-ofb -aria-256-cbc
-aria-256-cfb -aria-256-cfb1 -aria-256-cfb8
-aria-256-ctr -aria-256-ecb -aria-256-ofb
-aria128 -aria192 -aria256
...

```
seed@VM:~/.../Lab1$ openssl enc -aes-128-ecb -in pic_test.bmp -out pic_encrypted -k secret -pbkdf2
seed@VM:~/.../Lab1$ head -c 54 pic_test.bmp > header
seed@VM:~/.../Lab1$ tail -c +55 pic_encrypted > body
seed@VM:~/.../Lab1$ cat header body > pic_new.bmp
seed@VM:~/.../Lab1$ eog pic_test.bmp
seed@VM:~/.../Lab1$ eog pic_new.bmp
```



Origin



Encrypted in ECB mode

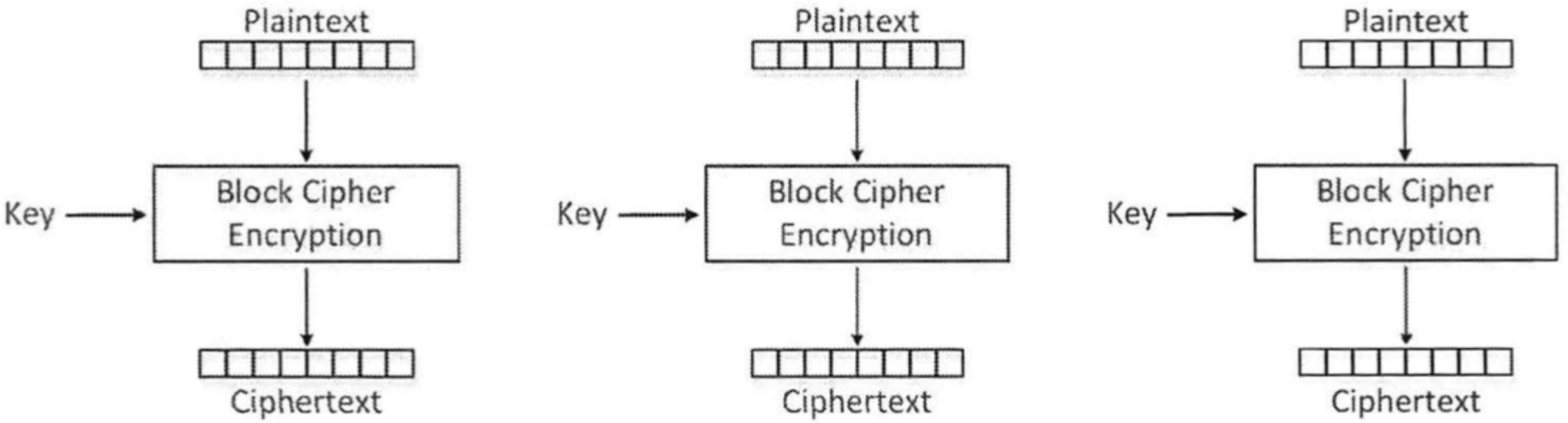
The simplest of the encryption modes is the **electronic codebook (ECB)** mode (named after conventional physical codebooks[19]). The message is divided into blocks, and each block is encrypted separately.

Key derivation from password algorithm PBKDF2: https://www.ietf.org/rfc/rfc2898.txt
Bitmap File Format: https://www.fileformat.info/format/bmp/egff.htm

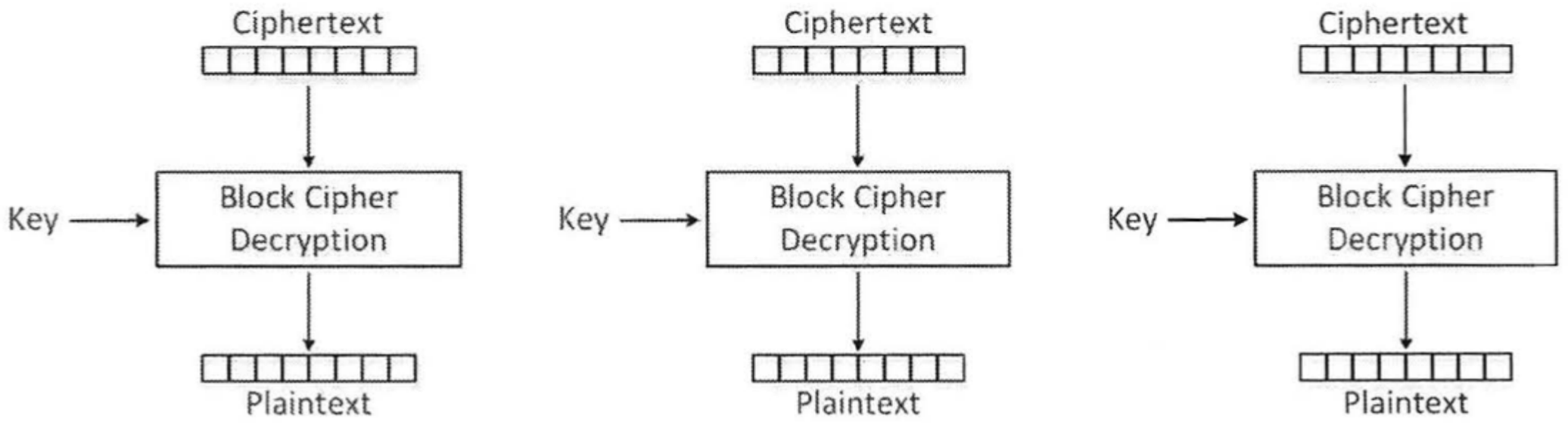
Encryption modes

- Must make the ciphertext blocks different even if their corresponding plaintext blocks are the same.
- A typical block cipher algorithm has two inputs: the plaintext block and the encryption key. Since the key is typically the same for all blocks, if the plaintext block is also the same, the output (ciphertext block) will obviously be the same. If we want to make the output different even for two identical blocks, we have to make one of the inputs different.
- There are many ways to make the inputs different → **encryption modes**

Electronic Code Book (ECB)

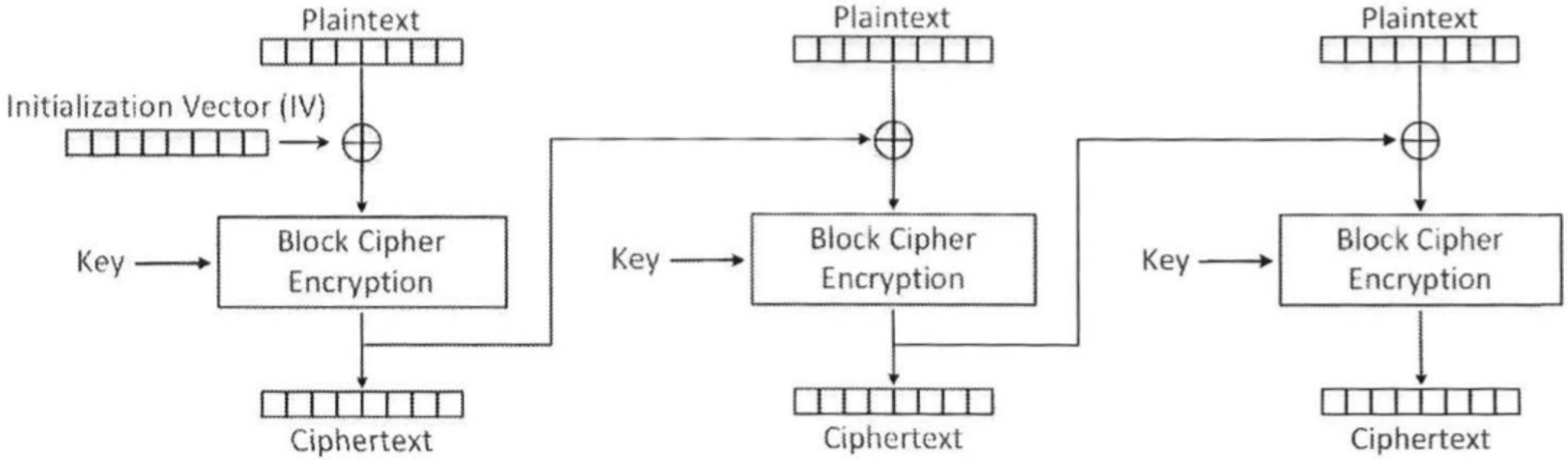


(a) Electronic Codebook (ECB) mode encryption

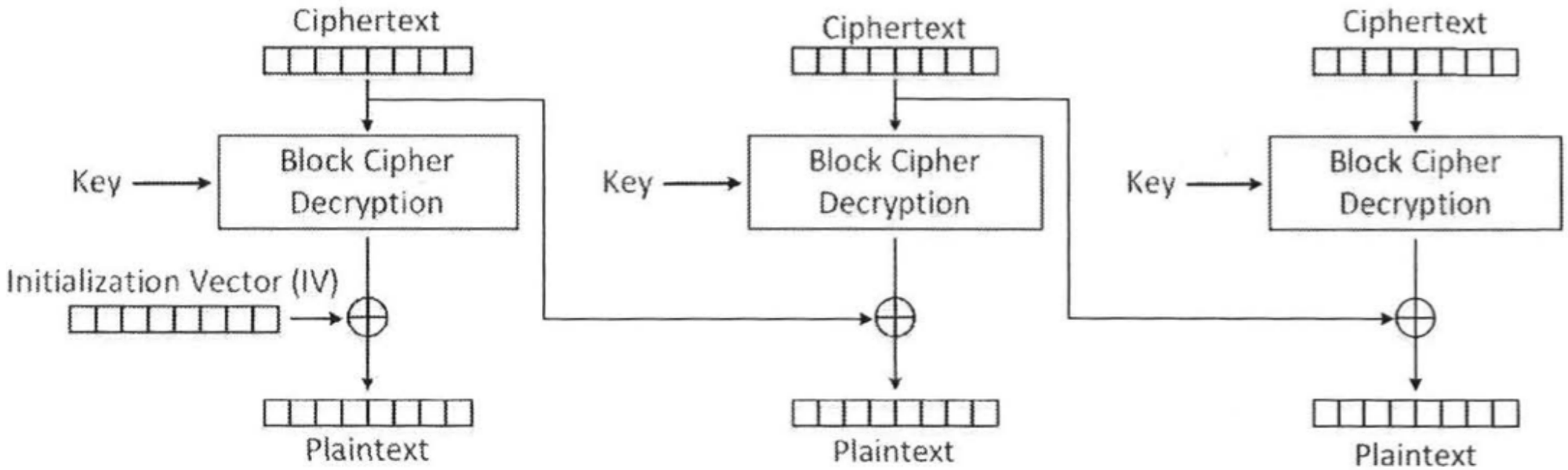


(b) Electronic Codebook (ECB) mode decryption

Cipher Block Chaining (CBC) Mode

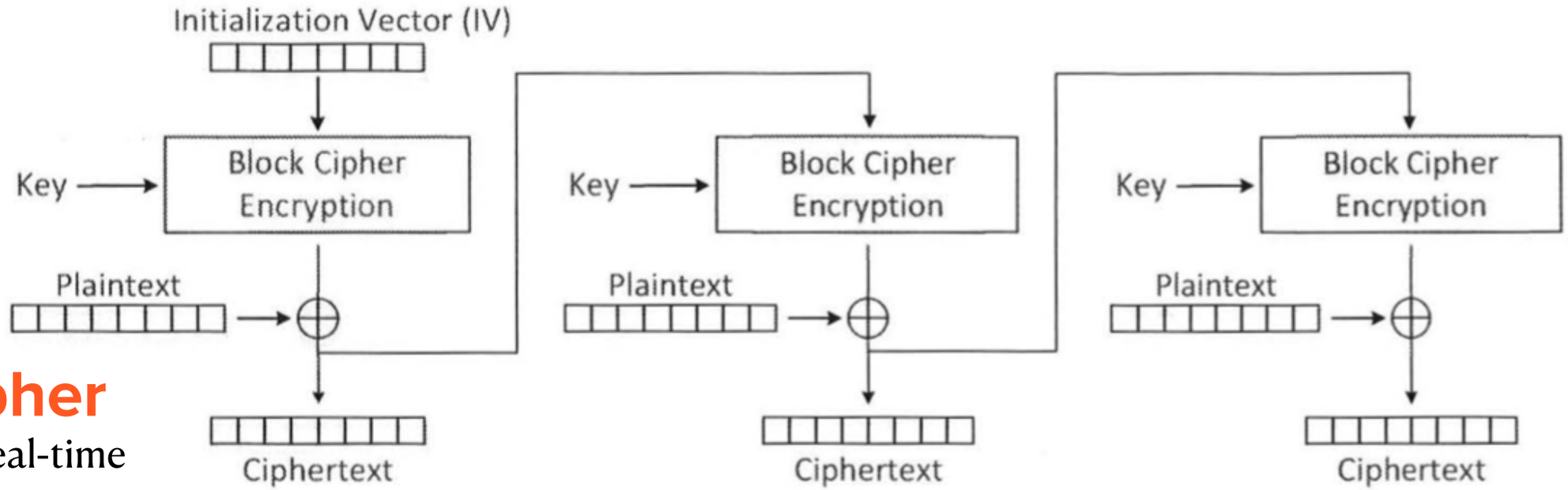


(a) Cipher Block Chaining (CBC) mode encryption

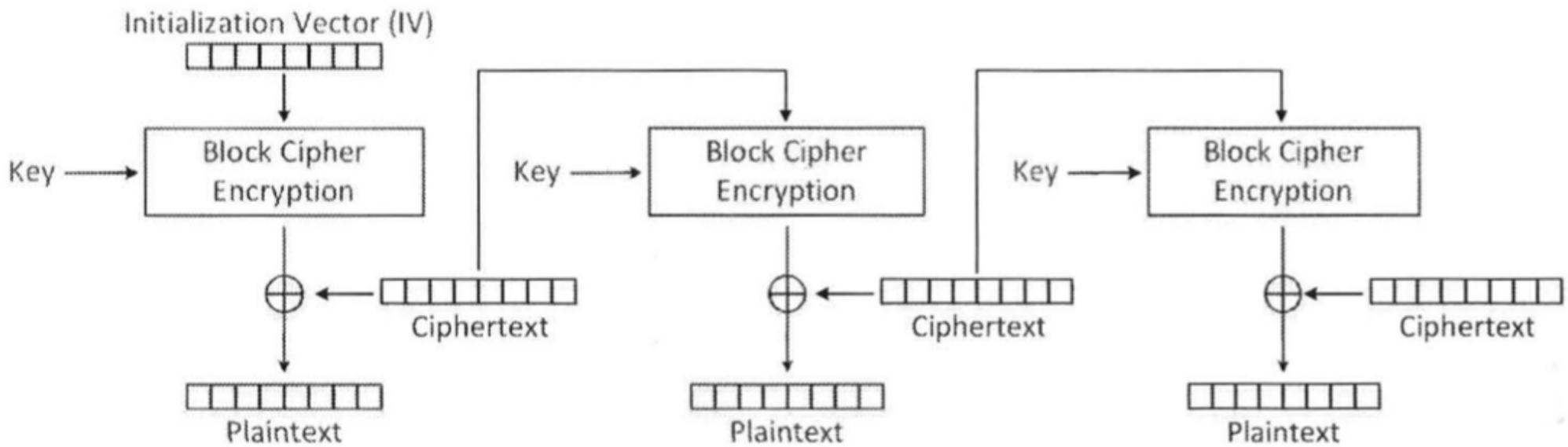


(b) Cipher Block Chaining (CBC) mode decryption

Cipher Feedback (CFB) Mode



(a) Cipher Feedback (CFB) mode encryption



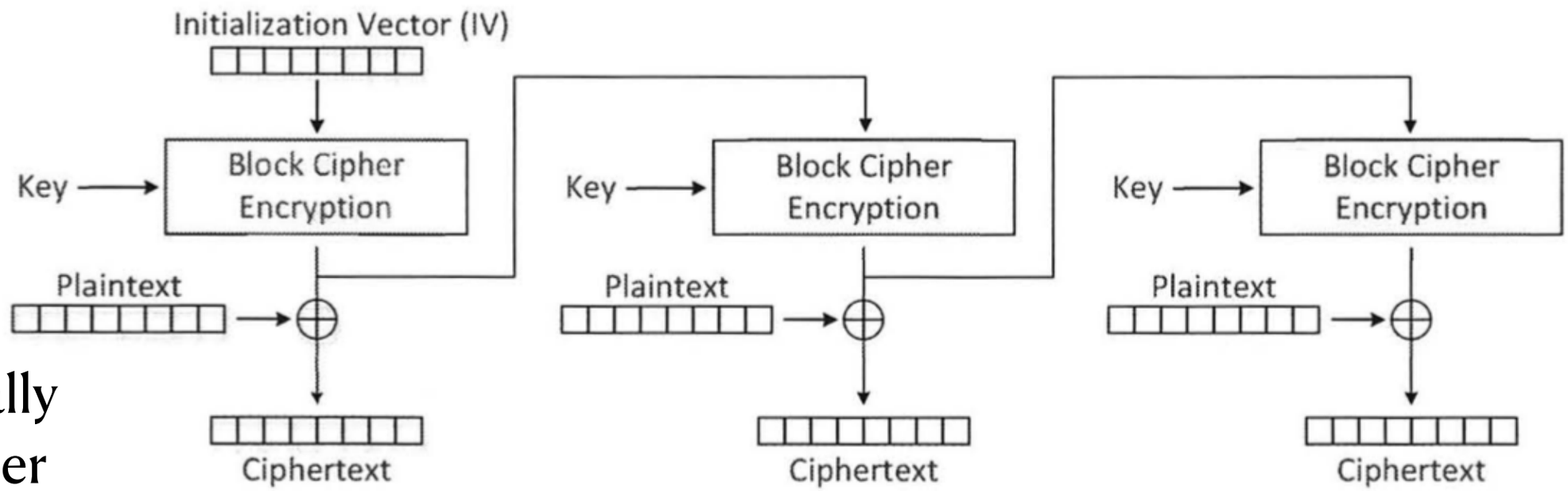
(b) Cipher Feedback (CFB) mode decryption

→ Stream Cipher

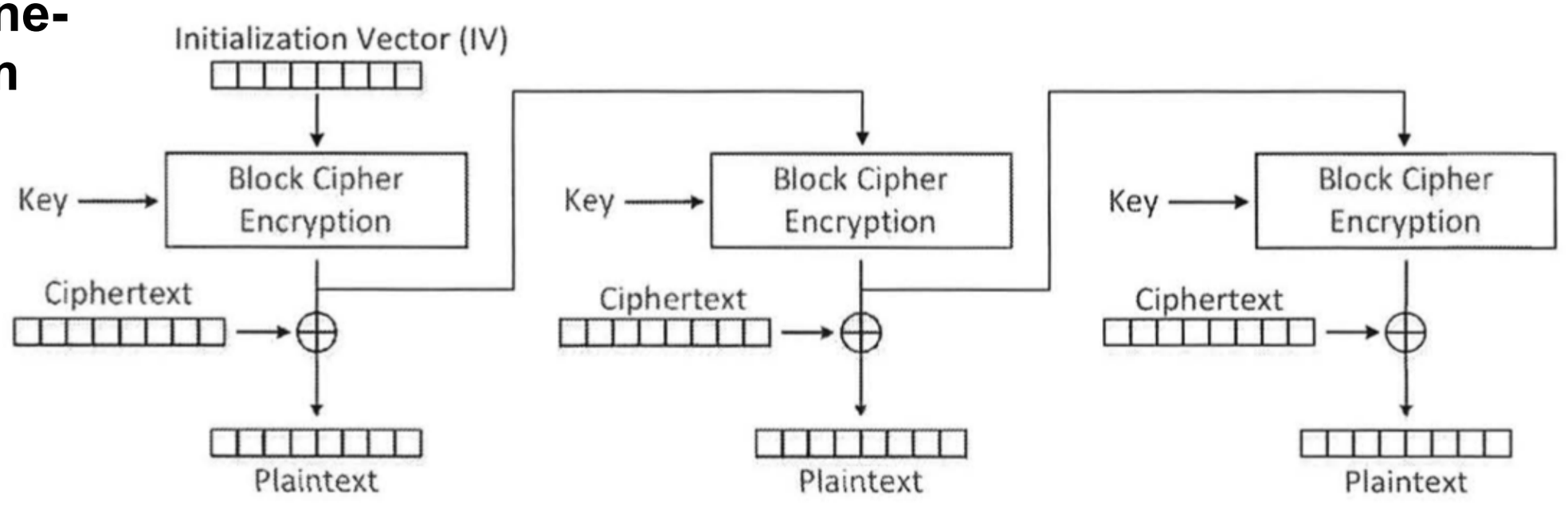
useful for encrypting real-time data, especially for situations where data generation is slow, i.e., it takes a while to fill a entire block.

Output Feedback (OFB) Mode

OFB mode basically uses a block cipher and a random IV to generate such a **one-time key stream**



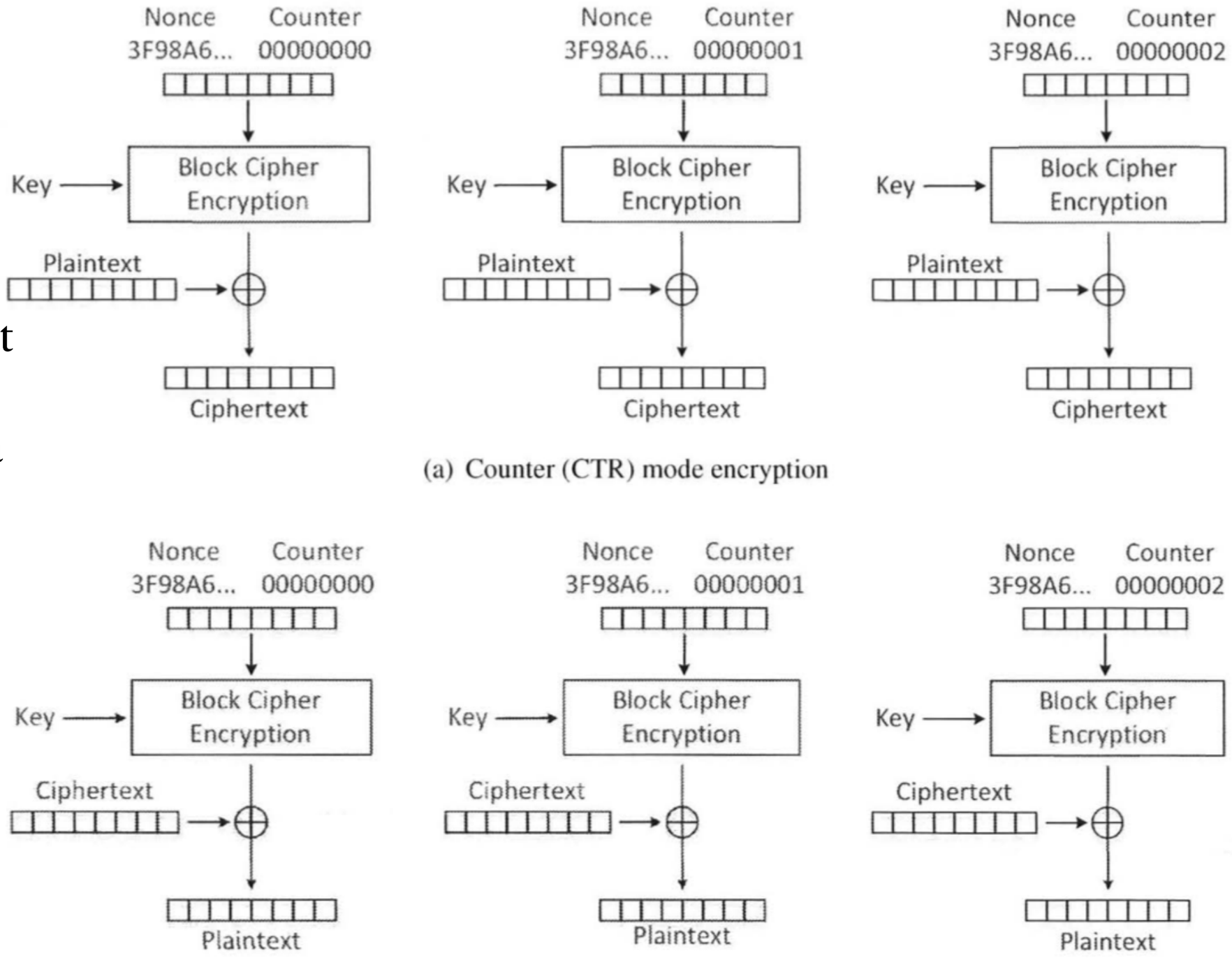
(a) Output Feedback (OFB) mode encryption



(b) Output Feedback (OFB) mode decryption

Counter (CTR) Mode

We do need to ensure that the key streams used for encrypting different data are different; **no key stream can be reused**



(a) Counter (CTR) mode encryption

(b) Counter (CTR) mode decryption